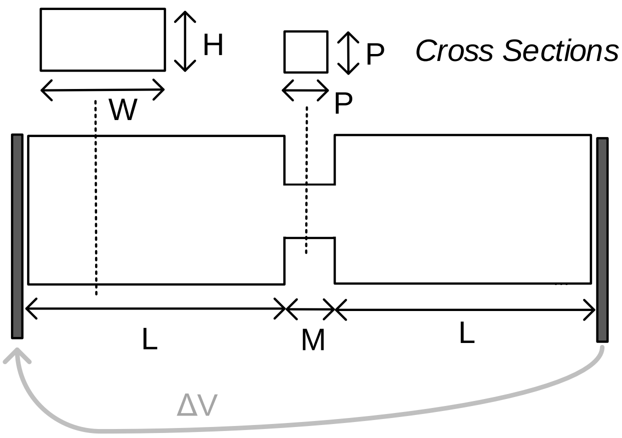
Politecnico di Milano

Biochip A.Y. 2018/2019 - M. Carminati

Exam of 13.02.2020

Test duration: 2 hours – No support material (textbooks, laptops, smartphones, notes) admitted.
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

Exercise 1
Consider a rigid microfluidic system for DNA electronic sequencing by means of a nanopore:



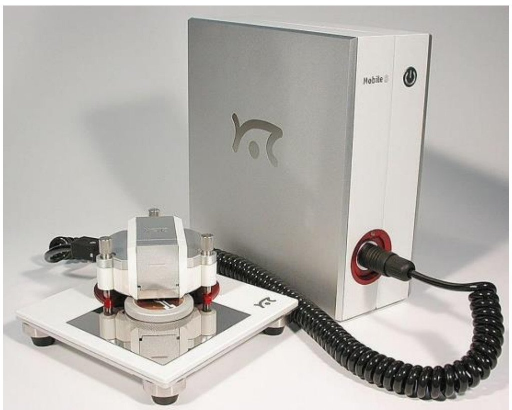
Data:

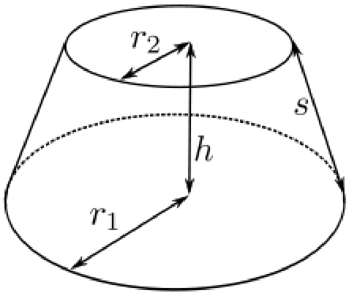
H = 20 μm
W = 100 μm
L = 1 mm
M = 1 μm

1atm = 101 kPa
$\eta_{water} = 10^{-3}$ Pa·s

a) Find the equivalent models of the system, from both the hydrodynamic and electrical points of view, obtaining the expression of each component of the model.
b) Choose the optimal size of P.
c) Analyze how both models of point (a) change if DNA molecules are suspended in blood, instead of PBS.
d) Analyze how both models of point (a) change with temperature.
e) Would it be possible to drive the flow of liquid through with pore with a reasonable (1atm) pressure difference?
f) Assuming a DNA mobility of 0.1 $mm^2/(V\ s)$, determine a proper voltage difference ΔV required to drive the translocation at a speed suitable for electrical sequencing.

Exercise 2
Now consider an Atomic Force Microscope (AFM) with a conductive cantilever. The tip can be modeled as a truncated cone (B):





(B) Tip model

Data:

r1 = 2 nm
r2 = 50 nm
h= 100 μm

ε_0 = 8.86e-12 F/m
ε_{r_water} = 80
δ_{PBS} = 1.5 S/m
C_{DL} = 0.1 pF/μm^2

a) Draw a detailed block scheme of a standard AFM and briefly explain its working principle in standard operating modes.
b) In case a current of 1 nA is to be measured, draw the scheme of a suitable current amplifier connected to the tip and size the components of the circuit (single power supply +5V).
c) Is the noise of the amplifier negligible with respect to the shot noise?
d) In case the impedance of a sample (placed between the tip and bottom holder) is to be measured, draw the scheme of the impedance measuring circuit.
e) In case the sample to be electrically measured is in liquid, find the equivalent impedance model between the two electrodes (tip and holder), both in PBS. Which parameters of the model can be quantitatively estimated (assuming the tip to be covered in gold)?
f) Now consider a three-electrode configuration, with the tip as Working Electrode and the holder as Counter Electrode. Draw the scheme of a potentiostatic circuit. Where should the Reference Electrode be placed?